

TIANJIN, China

WMO: 545270

Lat: 39.0812N

Lon: 117.0525E

Elev: 5

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-8.8	-25.5	0.4	-4.8	-23.1	0.5	-3.8	11.0	-2.9	9.8	-2.7	2.2	0	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	34.8	23.2	33.4	23.3	32.2	23.1	27.6	31.3	26.7	30.2	25.8	29.3	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.1	30.0	25.6	20.9	29.2	24.7	19.8	28.4	88.1	31.6	84.0	30.3	80.2	29.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	7.5	6.3	-12.8	37.3	2.3	1.6	-14.5	38.5	-15.8	39.4	-17.1	40.3	-18.7	41.4
			-14.0	28.5	2.1	0.9	-15.6	29.1	-16.8	29.6	-18.0	30.1	-19.6	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	-2.7	0.7	7.7	15.1	21.2	25.4	27.5	26.4	21.8	14.7	6.0	-0.7		
	DBStd	11.12	3.02	3.42	4.16	3.98	3.27	2.74	2.25	2.23	3.01	3.54	4.14	2.87		
	HDD10.0	1227	395	261	97	5	0	0	0	0	0	6	132	332		
	HDD18.3	2689	653	494	331	111	11	0	0	0	7	121	371	590		
	CDD10.0	2557	0	0	25	159	347	461	543	508	353	151	11	0		
	CDD18.3	978	0	0	1	15	99	211	285	249	110	8	0	0		
	CDH23.3	9459	0	0	7	152	894	2071	3137	2420	737	42	0	0		
	CDH26.7	3621	0	0	1	32	309	872	1327	884	191	5	0	0		
(14) Wind	WSAvg	2.6	2.5	2.6	3.2	3.4	3.2	2.6	2.3	2.0	2.1	2.3	2.3	2.3		
(15) Precipitation	PrecAvg	517	2	6	7	24	36	79	154	118	47	28	14	2		
	PrecMax	760	12	19	43	78	70	157	317	343	121	146	56	11		
	PrecMin	364	0	0	0	3	5	5	72	45	2	1	0	0		
	PrecStd	111	3	6	10	22	20	38	56	60	30	28	15	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	13.6	23.6	29.9	34.8	36.7	36.7	34.7	32.1	26.9	18.8	9.2		
		MCWB	1.7	5.6	12.1	16.2	18.9	20.8	24.8	25.4	21.6	16.8	10.4	3.0		
	2%	DB	5.4	10.9	20.1	27.0	32.1	34.7	35.0	33.2	30.2	24.3	16.3	7.1		
		MCWB	0.3	4.4	10.1	15.1	18.1	21.2	25.1	24.5	20.8	15.5	9.4	1.8		
	5%	DB	3.8	8.8	17.3	24.7	30.0	33.0	33.5	31.9	28.9	22.3	14.2	5.3		
		MCWB	-0.7	3.1	8.8	14.0	17.9	21.4	25.0	24.4	20.5	14.9	8.2	0.9		
	10%	DB	2.2	6.8	14.7	22.6	28.1	31.2	32.2	30.8	27.4	20.5	12.2	3.8		
		MCWB	-1.6	1.8	7.3	13.0	16.9	21.0	24.7	24.1	19.6	13.9	7.2	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	6.7	13.6	18.7	22.3	25.4	28.7	28.5	25.8	19.3	12.2	4.5		
		MCDB	6.4	11.2	21.4	26.3	29.7	31.1	33.2	31.8	29.2	23.1	16.3	7.6		
	2%	WB	0.9	5.1	11.5	17.0	20.9	24.4	27.9	27.4	23.6	17.8	10.5	2.8		
		MCDB	4.2	9.4	18.1	23.7	27.9	30.3	32.1	30.8	26.9	21.3	14.2	5.3		
	5%	WB	-0.3	3.7	9.6	15.7	19.8	23.5	27.0	26.5	22.4	16.6	9.2	1.7		
		MCDB	2.8	8.0	15.8	22.4	26.7	29.0	30.8	29.8	26.1	20.3	12.9	4.3		
	10%	WB	-1.1	2.3	7.9	14.3	18.7	22.6	26.2	25.7	21.3	15.3	8.0	0.7		
		MCDB	1.7	6.2	13.9	20.7	25.3	27.9	29.9	28.8	25.4	19.3	11.6	3.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	9.8	10.6	10.7	9.5	7.6	7.5	9.1	9.5	8.5	7.9		
		MCDBR	10.6	12.3	13.4	14.1	14.2	12.4	9.9	9.0	10.7	11.6	11.3	10.3		
	5% WB	MCWBR	6.8	7.2	6.2	5.5	4.7	3.8	3.3	3.2	3.6	4.8	5.9	6.3		
		MCWBR	9.6	11.4	12.6	12.1	11.5	9.2	7.6	7.0	8.1	9.0	9.9	8.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.469	0.546	0.565	0.629	0.674	0.808	0.807	0.760	0.685	0.654	0.535	0.484		
		taud	1.860	1.729	1.715	1.637	1.587	1.417	1.460	1.524	1.598	1.611	1.791	1.878		
	Ebn at Noon	Ebn at Noon	639	643	690	679	661	576	571	582	590	543	567	587		
		Edh at Noon	148	193	216	248	265	315	299	275	241	215	158	136		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.36	3.07	4.46	5.32	5.94	5.30	4.78	4.64	4.15	3.18	2.42	2.11		
	RadStd	RadStd	0.24	0.36	0.40	0.29	0.41	0.50	0.34	0.28	0.34	0.32	0.34	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	-2.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	+61

Nomenclature: See separate page